Interacting Multipoles of Rank 1 and 1

$$-1 \cdot T_{\alpha\beta}\mu_{\alpha}^{B}\mu_{\beta}^{A} = R_z^{-5} \cdot -1 \cdot \left[\begin{array}{c} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^B \cdot \mu_\beta^A \\ -1 \cdot R_z^2 \cdot \delta\left(\alpha, \beta\right) \cdot \mu_\alpha^B \cdot \mu_\beta^A \end{array} \right] \tag{1}$$

Simplify deltas:

$$-1 \cdot T_{\alpha\beta}\mu_{\alpha}^{B}\mu_{\beta}^{A} = R_z^{-5} \cdot -1 \cdot \left[\begin{array}{c} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^B \cdot \mu_\beta^A \\ -1 \cdot R_z^2 \cdot \delta\left(\alpha, \alpha\right) \cdot \mu_\alpha^B \cdot \mu_\alpha^A \end{array} \right] \tag{2a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-5} \cdot -1 \cdot \left[\begin{array}{c} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^B \cdot \mu_\beta^A \\ -1 \cdot R_z^2 \cdot \mu_\alpha^B \cdot \mu_\alpha^A \end{array} \right] \tag{2b}$$

Simplify R:

$$-1 \cdot T_{\alpha\beta}\mu_{\alpha}^{B}\mu_{\beta}^{A} = R_z^{-5} \cdot -1 \cdot \left[\begin{array}{c} +3 \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \mu_z^A \\ -1 \cdot R_z^2 \cdot \mu_\alpha^B \cdot \mu_\alpha^A \end{array} \right] \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-5} \cdot -1 \cdot \left[\begin{array}{c} -1 \cdot R_z^2 \cdot \mu_\alpha^B \cdot \mu_\alpha^A \end{array} \right] \tag{3b}$$

Simplify Dipoles:

$$-1 \cdot T_{\alpha\beta}\mu_{\alpha}^{B}\mu_{\beta}^{A} = R_z^{-5} \cdot -1 \cdot \left[\begin{array}{c} -1 \cdot R_z^2 \cdot \mu_y^B \cdot \mu_y^A \end{array} \right] \tag{4}$$

Combining everything:

$$-1 \cdot T_{\alpha\beta}\mu_{\alpha}^{B}\mu_{\beta}^{A} = \left[\begin{array}{c} +1 \cdot R_z^{-3} \cdot \mu_y^B \cdot \mu_y^A \end{array} \right] \tag{5}$$